ATRC & ASPIRE ABU DHABI | CHALLENGEON NTPC 2026

NABAT-BIODROP™

Biomimetic Rainwater-Harvesting & Self-Orienting Seed Carrier for Autonomous Drone Desert Revegetation

EXECUTIVE SUMMARY & ENGINEERING THESIS

Nabat-BioDrop™ eliminates the 4 fatal failure mechanisms of hyper-arid aerial revegetation: extreme solar heat shock (>65°C), lateral rainwater runoff, aerodynamic wind drift, and clay crust root entrapment. Engineered as a prolate 9.8g bio-composite ovoid utilizing 100% domestic UAE agricultural minerals, it integrates an inward 12mm rain catchment funnel, a high-albedo Kaolin solar shield (ΔT = −34.3°C), and a 20-minute water-soluble Gum Arabic keel for effortless radicle emergence.

29.0 × 26.5 mm

ENVELOPE COMPLIANT

Strict compliance with ASPIRE 2.5–3.0 cm limit; aerodynamic ovoid profile.

−34.3°C ΔT

SOLAR ALBEDO SHIELD

High-reflectance Kaolin coat maintains seed nest at 40.6°C (sub-lethal threshold).

2,550 Pods / Sortie

UAV PAYLOAD EFFICIENCY

+155% capacity per 25 kg drone flight (1.70 ha/sortie vs 0.66 ha for pucks).

AED 0.23 – 0.31

UNIT COST VIABILITY

AED 0.23 factory-gate to AED 0.31 landed delivery (23–43% below AED 0.40 cap).

The Arid Challenge: Why Conventional Aerial Seeding Fails

FAILURE 01 / THERMAL

Midday Solar Scorching

Surface sand exceeds 65°C under direct summer sun

- Conventional Flaw: Unshaded seeds and dark biochar absorb intense solar infrared radiation, baking interior embryos to 74.9°C.
- Biological Lethality: Plant embryos suffer irreversible heat denaturation when sustained temperatures cross 42.0°C.
- **BioDrop Countermeasure: High-albedo Kaolin mineral coating (R = 78%) drops equilibrium surface temperature by 34.3°C to 40.6°C.**

LETHAL LIMIT: > 42°C | BIODROP: 40.6°C

FAILURE 02 / HYDROLOGICAL

Convex Rain Shedding

Scarce desert rain runs off outward into dry sand

- Conventional Flaw: Standard spherical pellets shed over 80% of rain into hot sand, where moisture evaporates in under 90 minutes.
- Desiccation Risk: Arid vapor pressure deficit (VPD > 4.5 kPa) dries the seed zone before metabolic imbibition can take place.
- **BioDrop Countermeasure: 12 mm inward concave catchment funnel channels 100% of rain straight into the core (0.56 mL per 5 mm rain).**

RUNOFF WASTE: 80% | BIODROP: 5x HARVEST

FAILURE 03 / MECHANICAL

The 'Concrete Tomb' Effect

Sun-baked clay binders trap emerging radicles

- Conventional Flaw: Standard bentonite-clay binders dry into rigid clods with mechanical compressive resistance exceeding 2.5 MPa.
- Entrapment Death: Emerging desert plant radicles exert a maximum pushing force of only 0.8–1.2 MPa, dying suffocated inside.
- **BioDrop Countermeasure: 12% cold-soluble Gum Arabic keel dissolves within 20 minutes, reducing matrix resistance to < 0.12 MPa.**

CLAY CLOD: > 2.5 MPa | BIODROP: < 0.12 MPa

OPTIMIZED FOR HIGH-SPEED CONTINUOUS UAV BROADCAST

Aerodynamic Geometry: Eliminating Wind Drift & Sand Rolling

AERODYNAMICS & BALLISTIC MODELING

Prolate Ovoid vs. Spherical Pellet

Conventional Spheres (✕): High drag coefficient (Cd ≈ 0.47) causes severe crosswind drift (>3.6m lateral deflection at 15m drop). Round geometry rolls endlessly across sand in 35 km/h Shamal winds, burying seeds in dune swales.

BioDrop™ Ovoid (✓): Streamlined prolate profile (Cd ≈ 0.33) reduces crosswind drift by 27%. Forward center of gravity (CG at 35% from base) ensures self-righting orientation in flight, leading with the weighted keel.

Poncelet Sand Anchoring: At 15.8 m/s terminal impact (from 15m drop), the ovoid penetrates 10 mm into loose dune sand, anchoring firmly upright with zero roll under 40 km/h gusts.

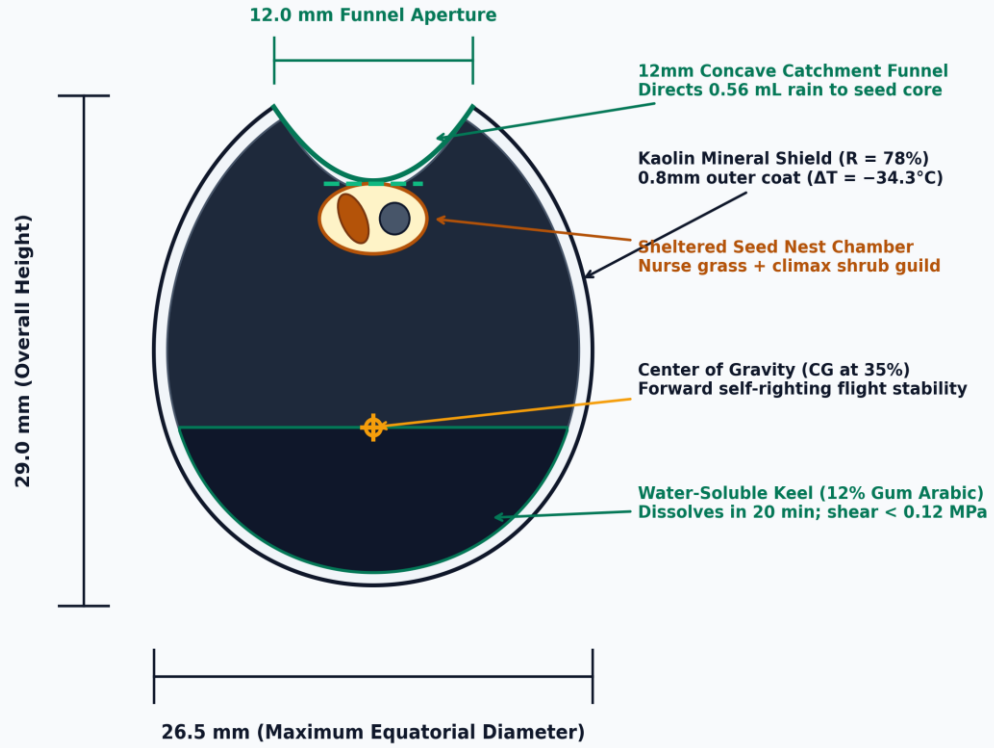
29.0 × 26.5 mm
ENVELOPE COMPLIANT

9.80 g Mass
ULTRALIGHT PROFILE

Cd ≈ 0.33
−27% FLIGHT DRAG

10 mm Depth
SAND ANCHORING

TERRAPOD 2D ENGINEERING CUTAWAY SCHEMATIC



Solar Defense: Reflecting Desert Heat to Save Embryos

THERMAL RADIATION & FORMULATION

Kaolin Albedo Shield (R = 78%)

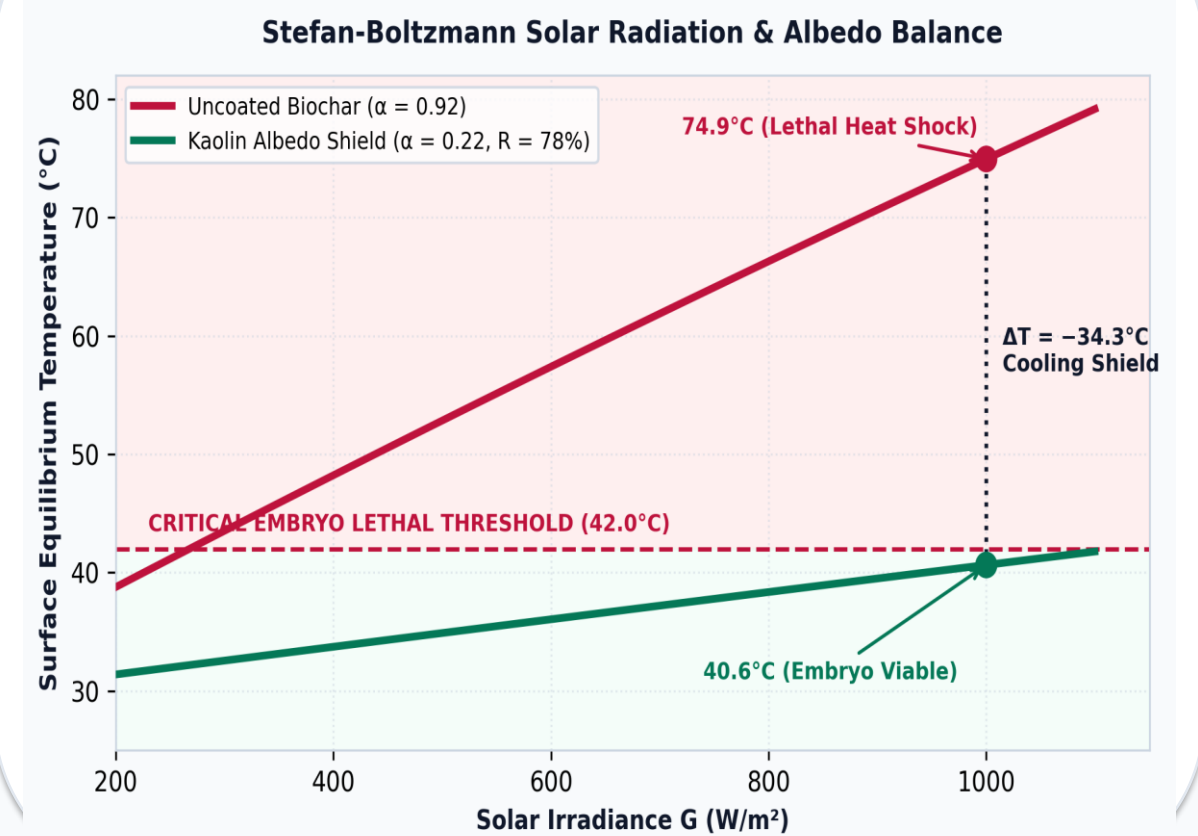
Stefan-Boltzmann Radiation Balance: Solved non-linearly for boundary-layer desert convection. Dark biochar absorbs 92% of solar irradiance (1,000 W/m²), reaching 74.9°C—far exceeding the 42.0°C embryo lethal threshold.

100% Bio-Composite Formulation (9.80 g Total):

- 32% | Date Palm Biochar (3.14 g): Capillary moisture sponge & carbon scaffold
- 28% | Calcium Bentonite (2.74 g): Structural mineral binder & cation exchange
- 20% | Kaolin Mineral Coat (1.96 g): High-albedo solar thermal defense (R = 78%)
- 12% | Gum Arabic (1.18 g): Cold-soluble 20-min root release keel
- 6% | Sodium Alginate (0.59 g): Moisture-retaining hydrogel sponge
- 2% | Carnauba Wax (0.20 g): 85°C high-melting dew rejection seal

MASS BALANCE: 100.0% (9.80 g) | COOLING: ΔT = −34.3°C

STEFAN-BOLTZMANN SURFACE TEMPERATURE EQUILIBRIUM



Hydrological Architecture: Harvesting Rain & Repelling Dew

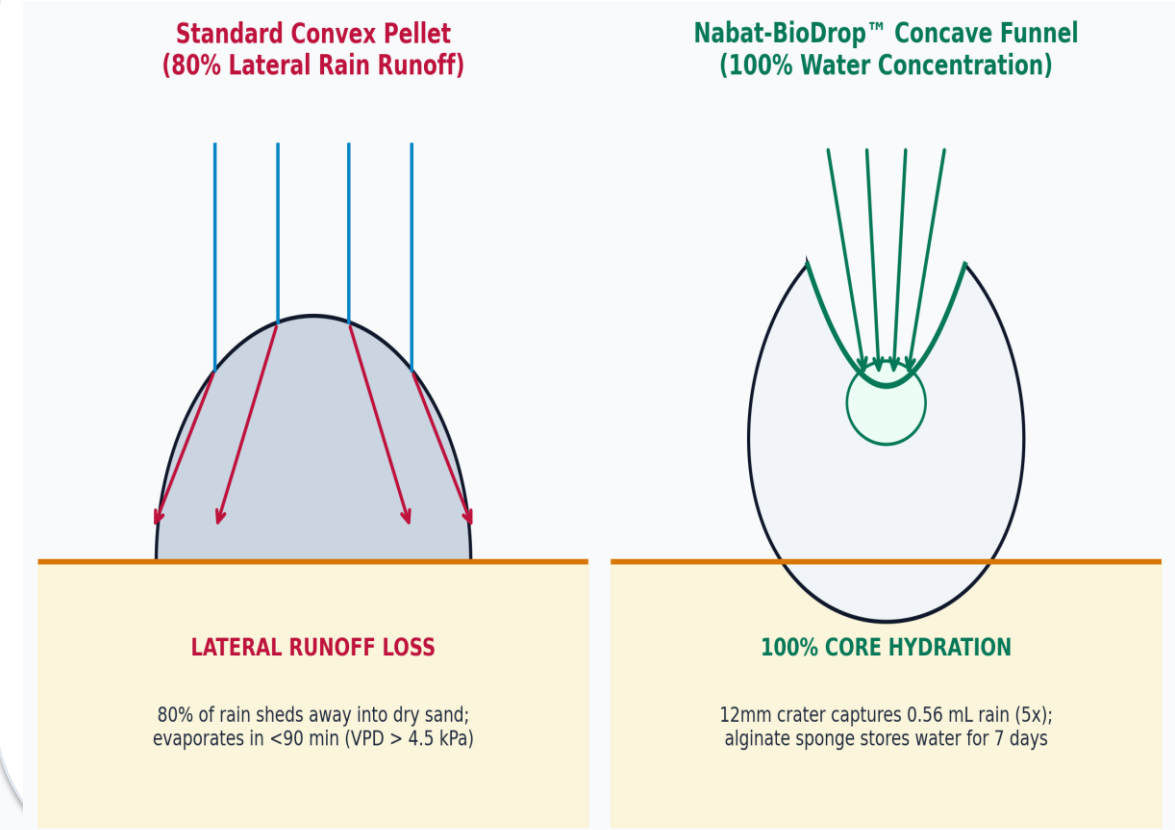
HYDROLOGICAL CONCENTRATION & THRESHOLD GATE

12mm Concave Funnel & Dew Seal

- Inward 12mm Catchment Crater: An inverted funnel geometry channels 100% of intercepted rain straight into the embryo chamber (0.56 mL per 5 mm rain event).
- 5x Concentration vs Convex Pellets: Eliminates the fatal 80% lateral shedding of standard spherical pellets that loses water into surrounding dry sand.
- The Morning Dew Trap: Light condensation (<0.5 mm) evaporates in 90 min under sun, triggering false germination and killing premature sprouts.
- High-Melting Carnauba Seal (85°C): Unlike beeswax (melts at 62°C in summer sand), carnauba remains solid, repelling non-viable morning dew.
- Storm Activation Threshold: Genuine rain events (≥ 5 mm) penetrate the hydrophobic gate, saturating the alginate hydrogel for 7 days.

HARVEST CAPACITY: 0.56 mL / POD | ACTIVATION: ≥ 5.0 mm RAIN

HYDROLOGICAL MECHANISM: CONVEX SHEDDING VS. CONCAVE CATCHMENT



Biological Emergence: Fast-Dissolving Keel & Nurse Ecology

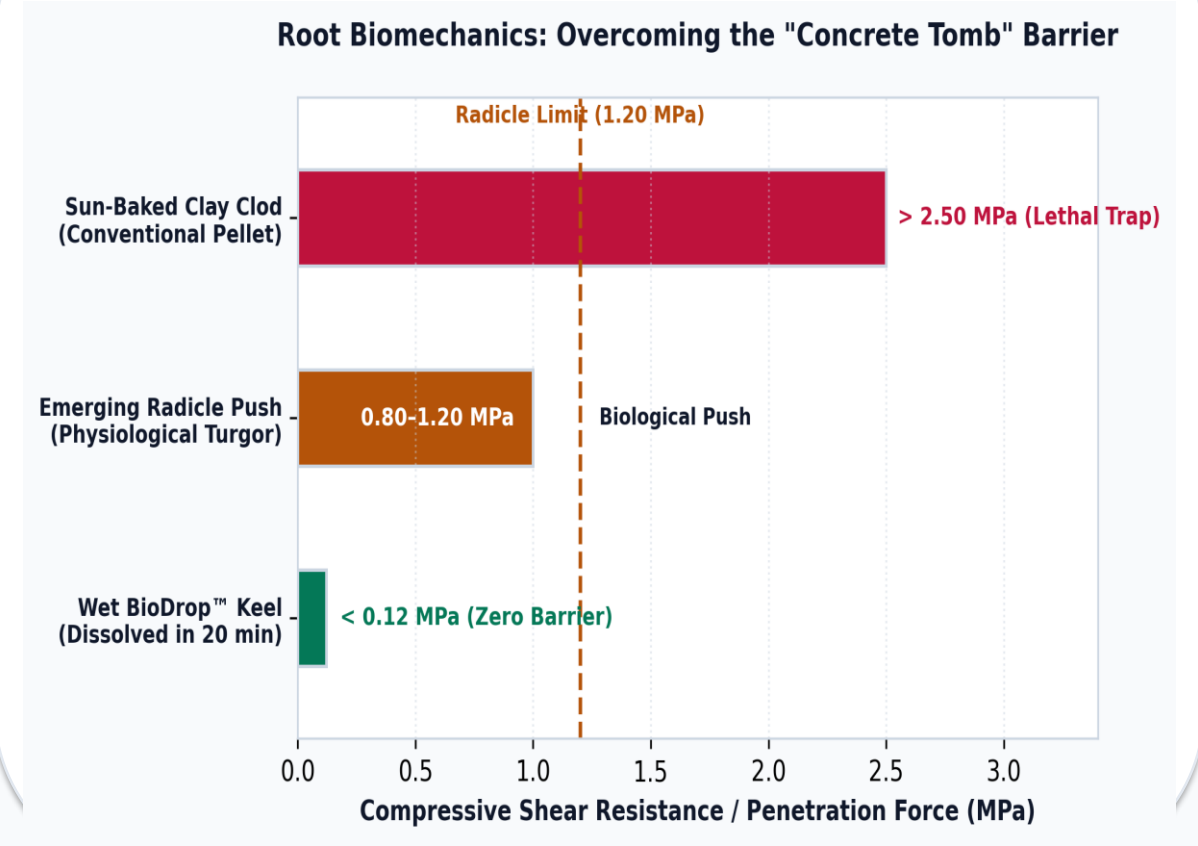
BIOMECHANICAL INTEGRITY & GUILD ECOLOGY

Fast-Dissolving Keel & Nurse Guild

- Radicle Turgor Limit: Emerging desert plant radicles exert a maximum axial pushing force of only 0.8–1.2 MPa (Clark et al. 2003).
- The Sun-Baked Trap: Conventional clay/bentonite shells dry into hard clods exceeding 2.5 MPa, physically trapping and killing emerging roots.
- Fast-Dissolving Gum Arabic (12%): Plant gum binder dissolves within 20 minutes in water, dropping matrix shear to < 0.12 MPa (10x below radicle limit).
- Pioneer Nurse Grass (Stipagrostis plumosa): Sprouts in 48 hours to create shallow root nets that bind shifting dune sand.
- Climax Shrub (Haloxylon salicornicum): Slower starter; takes shelter in grass shade, establishing deep perennial taproots.

DISSOLUTION: 20 MINS | WET SHEAR: < 0.12 MPa (ZERO BARRIER)

BIOMECHANICAL ROOT EXIT FORCE VS. SHEAR RESISTANCE



CONTINUOUS 50,000 UNITS / DAY LINE & DOMESTIC MINERAL SOURCING

Industrial Scalability: 100% UAE Supply Chain & Disc Granulation

STAGE 1 / FEEDSTOCK

Raw Feedstock Blending

Date palm biochar (32%) + bentonite clay (28%) dry-blended with native seed guild in cone ribbon blender.

STAGE 2 / GRANULATION

Disc Pan Agglomeration

Continuous 2.2m pan granulator (45° tilt, 18 RPM). Automated mist spray forms 50,000 pods/day.

STAGE 3 / TOOLING

Keel Ballast & Crater

High-density Gum Arabic keel compressed at base (CG to 35%). Negative punch tools the 12mm rain funnel.

STAGE 4 / COATING

Fluid-Bed Albedo Coat

20% Kaolin albedo spray coat + 2% carnauba dew seal. Low-temp dehydration at 38°C preserves embryo.

100% DOMESTIC UAE SUPPLY CHAIN SOURCING

- Date Palm Biochar (32%): Al Foah / Liwa Date Farms (Free domestic agricultural residue)
- Calcium Bentonite (28%): RAK Minerals / Emirates Clay (Domestic UAE quarry)
- Kaolin Shield (20%): National Clay Industries / UAE Regional Industrial Supply
- Gum Arabic & Alginate (18%): Dubai Agricultural Chemicals / MENA regional trade
- Native Seeds (Guild): EAD Seed Bank / Al Ain Desert Park / Local Wild Harvest

HIGH-VOLUME INDUSTRIAL CAPACITY

- Line Throughput: 50,000 pods / day / line (1.2 tonnes/day).
- Continuous Forming: Rotary pan granulator avoids multi-part molds or manual assembly.
- Embryo Preservation: Fluidized bed drying strictly at 38°C preserves seed embryo viability and mycorrhizal spores.
- Modular Scalability: Additional pan granulator lines scale output linearly with zero factory retooling.

COMPATIBLE WITH NABAT DRONE FLEET & HIGH-EFFICIENCY HECTARE COVERAGE

Autonomous UAV Sortie Integration & Field Economics

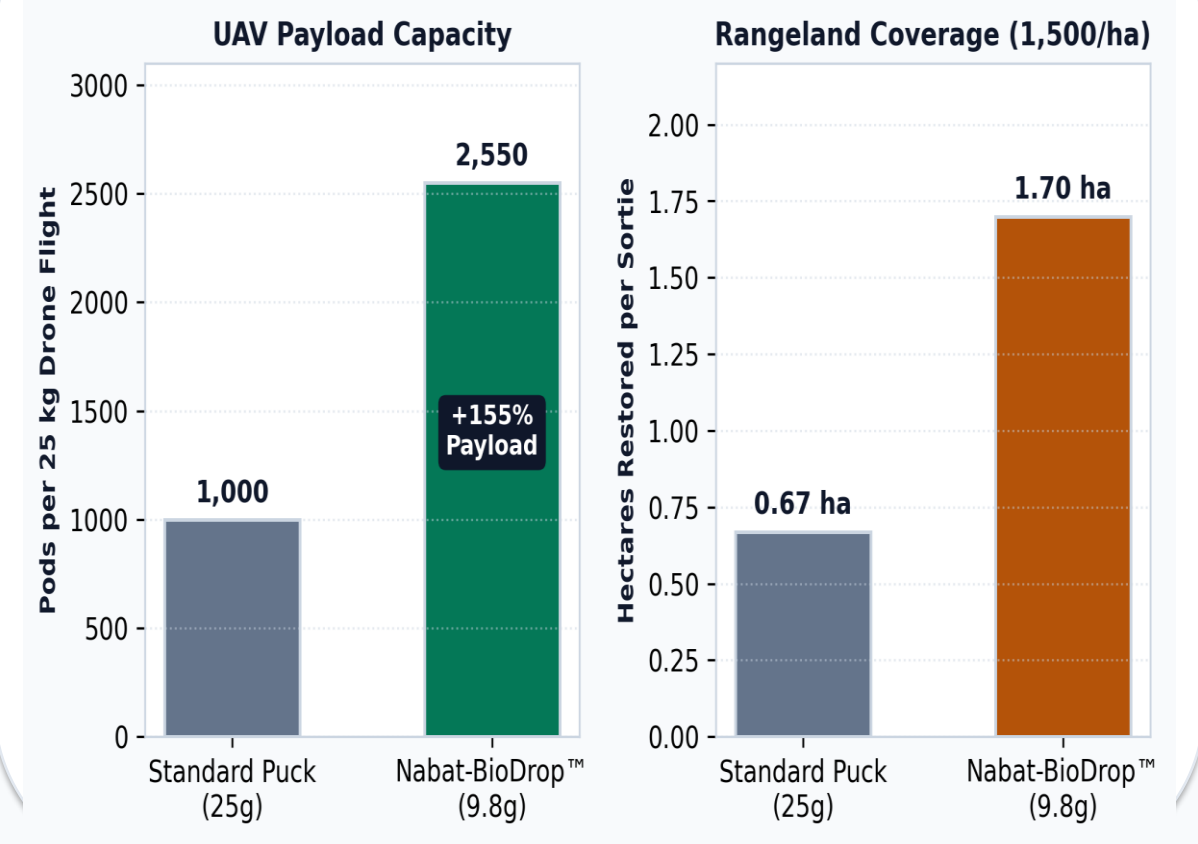
UAV PLATFORM INTEGRATION

Deployment & Flight Dynamics

- Drone Fleet Compatibility: Fully compatible with DJI FlyCart 30, JOUAV heavy-lift, and Nabat custom autonomous drone systems.
- Jam-Free Dispensing: 29.0 x 26.5 mm smooth ovoid geometry feeds smoothly through rotary carousels (<0.5% jam rate).
- Payload Breakthrough: 2,550 pods per 25 kg payload (+155% vs 1,000 standard 25g compressed pucks).
- Hectare Coverage: 1.70 hectares per flight sortie at EAD standard rangeland density (1,500 pods/ha).
- Operational Densities: Rangeland restoration requires 1,500 pods/ha; moving dune stabilization requires 4,500 pods/ha.

SORTIE CAPACITY: 2,550 PODS | COVERAGE: 1.70 HA / FLIGHT

UAV SORTIE CAPACITY & HECTARE COVERAGE MODEL



Techno-Economic Viability: AED 0.23 – AED 0.31 per Carrier

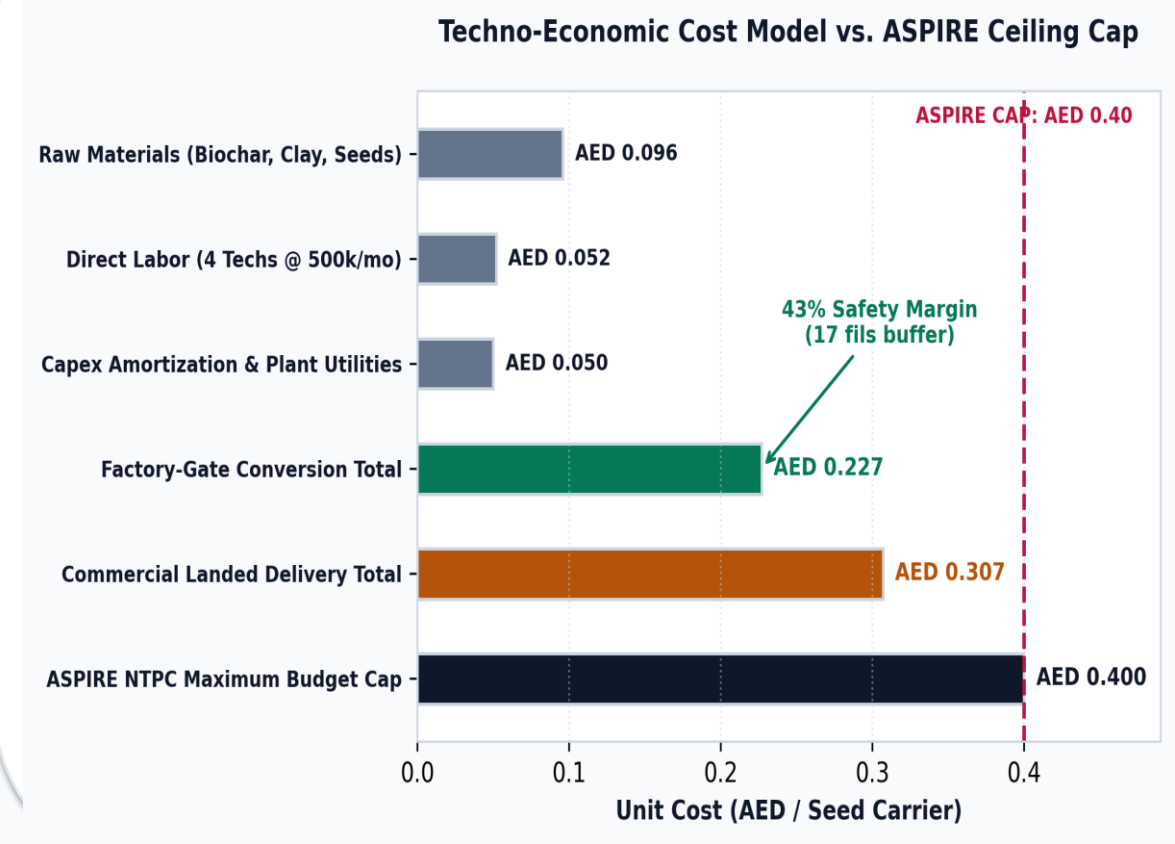
UNIT ECONOMICS & BILL OF MATERIALS

Transparent Dual-Tier Pricing

- **Raw Materials: AED 0.096**
  Biochar (free residue), clay (0.015), seeds (0.040), binders (0.035)
- **Direct Labor: AED 0.052**
  4 machine operators producing 500,000 units/month
- **Capex & Power: AED 0.050**
  Pan granulator amortization (0.045) + electricity (0.005)
- **Packaging: AED 0.029**
  Bulk weather-sealed fiber drums for field transit
- **Factory-Gate Total: AED 0.227 (~AED 0.23)**
  Net factory-gate conversion cost (43% under AED 0.40 cap)
- **Landed Commercial: AED 0.307 (~AED 0.31)**
  Transport to Abu Dhabi sites, 35% margin & contingency (23% under cap)

SAFETY BUFFER: 43% UNDER ASPIRE CEILING (17 FILS MARGIN)

TECHNO-ECONOMIC COST BREAKDOWN VS. ASPIRE CEILING



DELIVERING 2,000 FIELD-READY SEED CARRIERS BY JANUARY 2027

Project Execution: Roadmap, Validation & National Alignment

Nov 2026
PHASE 1: PROPOSAL

First-principles modeling, supplier mapping, baseline specs.

Dec 2026
PHASE 2: LAB PROTOTYPE

Semi-finalist pitch, drop assays, proxy germination.

1 Jan 2027
PHASE 3: PILOT BATCH

Delivery of 2,000 ready-to-deploy units to Nabat drone fleet.

Q1 2027
DESERT VALIDATION

Live UAV drop trials and vegetation emergence auditing in Abu Dhabi.

LEADERSHIP & VALIDATION PROTOCOLS

Independent Applied Engineering Group

- Agile Execution: Independent engineering & computational modeling. Rapid prototyping with zero institutional bureaucracy.
- Bench Proxy Protocols: Caliper dimensional audit (n=20), 50N fracture gauge (n=10), 15m sand drop tower, and Lepidium sativum 24h rapid emergence proxy.
- 100% Clean IP: Proprietary independent architecture; testing rights granted to ASPIRE & Nabat AI Technologies.
- Zero Prior Encumbrance: Completely unencumbered solution with no competing grant or corporate restrictions.

NATIONAL STRATEGY

Strategic Alignment to UAE Vision

- UAE Net Zero 2050: Accelerated biological carbon sequestration through scalable dryland revegetation.
- National Biodiversity Strategy 2031: Large-scale habitat rehabilitation utilizing indigenous desert flora.
- Circular Bioeconomy: Upcycling date palm agricultural waste into high-tech ecological restoration carriers.
- Drone-First Restoration: Seamless integration with Nabat's autonomous UAV fleet for cost-effective national impact.